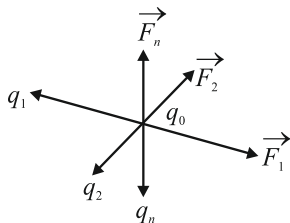


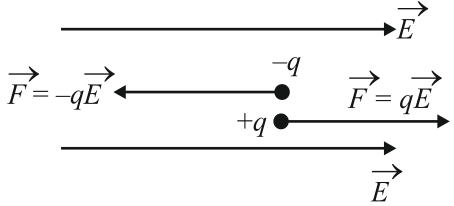
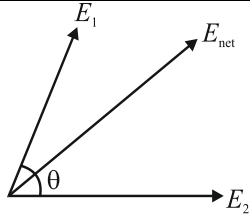
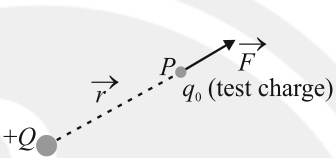
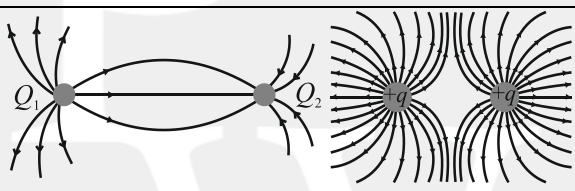


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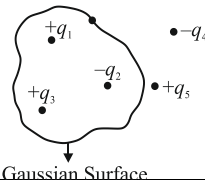
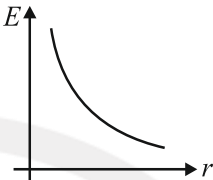
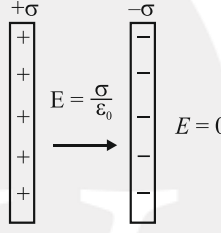
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Electric Charges and Fields

Sr. No.	Concept	Formulas	Other information
1.	Electric charge (Q)	Quantization of charge: $Q = \pm ne$ Q = Total charge $n = 1, 2, 3 \dots$ $e = 1.6 \times 10^{-19} \text{C}$ The minimum charge possible is $1.6 \times 10^{-19} \text{C}$ 1 coulomb = 3×10^9 esu	Q = total electric charge n = number of electrons. e = charge of one electron. esu is the electrostatic unit of charge
2.	Additive property of electric charge & redistribution of charges	Total charge (Q') if system contains n charges $q_1, q_2, q_3 \dots q_n$, then $Q' = \sum q_n = q_1 + q_2 + q_3 + \dots q_n$ $Q = \frac{Q_1 + Q_2}{2}$ Q = Charge on each shell after redistribution	r = radius of spheres Q_1, Q_2 = charge on the two spheres
3.	Coulomb's law	 The force between two charges $F_e = \frac{1}{4\pi\epsilon_0} \frac{ q_1 q_2 }{r^2}$ Vector form $\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}$ In medium having relative permittivity (ϵ_r) $F = \frac{1}{4\pi\epsilon_0 \epsilon_r} \frac{q_1 q_2}{r^2}$	r = distance between two charges q_1 and q_2 . $k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \frac{\text{N} \times \text{m}^2}{\text{C}^2}$ The permittivity of free space $\epsilon_0 = 8.854 \times 10^{-12} \text{C}^2/\text{N} \cdot \text{m}^2$
4.	Superposition principle	Vector sum of all the individual forces $\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \dots \vec{F}_{1N}$ or $\vec{F} = \frac{kq_0 q_1}{r_1^2} \hat{r}_1 + \frac{kq_0 q_2}{r_2^2} \hat{r}_2 + \dots + \frac{kq_0 q_i}{r_i^2} \hat{r}_i$ $+ \dots \frac{kq_0 q_n}{r_n^2} \hat{r}_n$	
5.	Electric field (E)	$\vec{E} = \frac{\vec{F}}{q_0}$ SI unit: N/C or V/m. Dimensional formula $[M^1 L^1 T^{-3} A^{-1}]$.	$\vec{E}$ = electric field vector q_0 = electric charge

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6.	Force on electric charge (F)		$+q, -q$ = electric charges.
7.	Superposition of electric field	 General rule $E_{Net} = \sqrt{E_1^2 + E_2^2 + 2E_1E_2 \cos \theta}$	θ = angle between E_1 and E_2
8.	Electric field due to a point charge	$\vec{E} = \frac{\vec{F}}{q_0} = \frac{kQ}{r^2} \hat{r}$ $\vec{E} = \frac{kQ}{r^3} \vec{r}$ 	Here, $k = \frac{1}{4\pi\epsilon_0}$
9.	Electric field lines	 Tangent drawn at any point on the electric field line gives the direction of force.	$ Q \propto$ number of field lines Here, $ Q_1 > Q_2 $
10.	Charge density	Volume charge density, $\rho = \frac{dq}{dV} \text{ Cm}^{-3}$ Surface charge density, $\sigma = \frac{dq}{dS} \text{ Cm}^{-2}$ Linear charge density, $\lambda = \frac{dq}{dL} \text{ Cm}^{-1}$	dq = amount of charge present in a small surface or volume.
11.	Electrical field due to a finite length of the wire	$E_x = E_{\perp} = \frac{k\lambda}{r} (\sin \theta_1 + \sin \theta_2), \quad \tan \alpha = \frac{E_y}{E_x}$ $E_y = E_{\parallel} = \frac{k\lambda}{r} (\cos \theta_1 - \cos \theta_2), \quad E = \sqrt{E_x^2 + E_y^2}$ E_x and E_y are the electric fields along the x and y axis.	For infinite wire, $\theta_1 = \theta_2 = 90^\circ$. λ = linear charge density r = distance from the wire.
12.	Electric field due to a uniformly charged arc at centre	$E = \frac{2k\lambda}{R} \sin\left(\frac{\theta}{2}\right), \quad \theta \rightarrow \text{angle of arc}$	R = radius of the arc. λ = linear charge density

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13.	Electric field on an axial point of ring	$E_x = \left(\frac{1}{4\pi\epsilon_0} \right) \frac{qx}{(x^2 + R^2)^{\frac{3}{2}}}$ $E_x = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{x^2} \text{ for } x \gg R, \text{ i.e. when the point is much farther from the ring.}$	E_x = electric field at distance x from the centre of uniformly charged ring. q = total charge on its axis.
14.	Electric dipole moment.	$\vec{p} = q\vec{\ell}$ It is a vector quantity whose direction is from ($-q$) to ($+q$) SI unit $\rightarrow$ C-m	ℓ = the separation between two charges. p = the dipole moment of the system
15.	Electric field due to a dipole	$E_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \cdot \frac{2pr}{(r^2 - a^2)^2} \approx \frac{1}{4\pi\epsilon_0} \cdot \frac{2p}{r^3} \text{ for } r \gg a.$ $E_{\text{equa}} = \frac{1}{4\pi\epsilon_0} \cdot \frac{p}{(r^2 + a^2)^{\frac{3}{2}}} \approx \frac{1}{4\pi\epsilon_0} \cdot \frac{p}{r^3} \text{ for } r \gg a.$	E_{axial} = The electric field on the axis at distance r from mid-point. E_{equa} = Electric field at an equatorial point of a dipole at a distance r from charge. $2a$ = dipole length.
16.	Electric field due to a dipole at general position	Net electric field at that point will be $E = \sqrt{E_1^2 + E_2^2}$ $E = \frac{kp}{r^3} \sqrt{(2\cos\theta)^2 + (\sin\theta)^2} = \frac{kp}{r^3} \sqrt{3\cos^2\theta + 1}$	r = distance between the test point and midpoint of the electric dipole. θ = the angle between the dipole moment and line joining the test point and midpoint of the electric dipole. p = dipole moment.
17.	Torque on a dipole placed in external electric field	Magnitude of torque = $qE \times 2a \sin\theta$ $= 2qaE \sin\theta$ $\tau = pE \sin\theta \quad [\because p = (2a)q]$ or $\vec{\tau} = \vec{p} \times \vec{E}$	$2a$ = distance between the charges. E = electric field
18.	Work done in rotating a dipole	$W = \int_{\theta_1}^{\theta_2} pE \sin\theta d\theta$ $W_{\text{external force}} = pE(\cos\theta_1 - \cos\theta_2)$	W = total work done by external forces in rotating a dipole from θ_1 to θ_2
19.	Electric flux (ϕ)	$\phi_E = \int_S \vec{E} \cdot d\vec{S} \Rightarrow \phi_E = \int_S E dS \cos\theta = E \cos\theta \int_S dS$	θ = the small angle between E and dS . dS = small area

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20.	Gauss's Theorem	$\phi_{\text{net}} = \oint_S \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$ To understand this concept consider a closed surface as shown $\phi_{\text{net}} = \frac{(q_1 - q_2 + q_3)}{\epsilon_0}$ 	ϕ_{net} = net electric flux passing through any closed imaginary surface
21.	Electric field due to an infinitely long straight uniformly charged wire (E)	$E = \frac{\lambda}{2\pi\epsilon_0 r} \quad E \propto \frac{1}{r}$ So, E-r graph is a rectangular hyperbola as shown in figure for infinite plane sheet. 	λ = linear charge density (charge per unit length), r = located distance from the line charge
22.	Electric field intensity due to a uniformly charged infinite plane sheet	For infinite plane sheet: Using Gauss's law, $\phi_E = \frac{q}{\epsilon_0}$ $2EA = \frac{\sigma A}{\epsilon_0} \Rightarrow E = \frac{\sigma}{2\epsilon_0}$ For parallel plane sheets of opposite surface charge densities, $E = \frac{\sigma}{\epsilon_0}$ (between two plates) $E = 0$ (outside the plates) For a thick conducting sheet $E = \frac{\sigma}{\epsilon_0}$ 	σ = Surface charge density Permittivity of free space $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/\text{N-m}^2$
23.	Electric field of a thin spherical shell	$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} \text{ For } r > R \text{ (Outside points)}$ $E = 0 \text{ For } r < R \text{ (Inside points)}$ $E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{R^2} \text{ For } r = R \text{ (At the surface)}$ where $q = 4\pi R^2 \sigma$	R = radius of spherical shell σ = surface charge density of the shell.
24.	Electric field of a uniformly charged solid sphere	$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} \text{ For } r > R \text{ (Outside points)}$ $E = \frac{1}{4\pi\epsilon_0} \cdot \frac{qr}{R^3} \text{ For } r < R \text{ (Inside points)}$ $E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{R^2} \text{ } r = R \text{ (At the surface)}$ Where $q = \frac{4}{3}\pi R^3 \rho$	ρ = uniform volume charge density R = radius of the sphere.